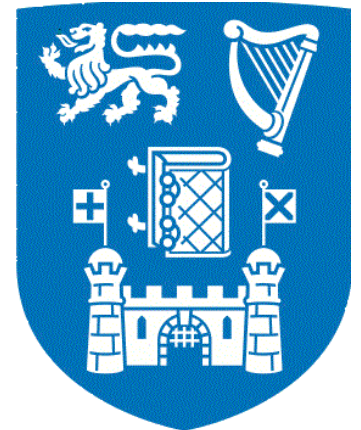


Formal Verification CSU44004/CSU55004

Lecture 5: Propositional Logic

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Natural Deductions Proofs

❖ Modus Ponens ($\rightarrow$ e)

❖ Example 2:

❖ Premises: $p \rightarrow q$

$q \rightarrow r$

p

❖ Goal: Show r .

❖ Example 3:

❖ Premises: $p \wedge q$

$p \rightarrow r$

❖ Goal: Show r .

1	$p \rightarrow q$	Premise
2	$q \rightarrow r$	Premise
3	p	Premise
4	q	$\rightarrow$ e from (1,3)
5	r	$\rightarrow$ e from (2,4)

1	$p \wedge q$	Premise
2	$p \rightarrow r$	Premise
3	p	$\wedge e_1$ from (1)
4	r	$\rightarrow$ e from (2,3)

Natural Deductions Proofs

❖ Modus Tollens (MT)

❖ If A implies B, and B is false, then A must also be false.

❖ A: it rained

❖ B: streets are wet.

❖ Premises: $A \rightarrow B$: if it rained then the street is wet.

❖ The street is not wet. ($\neg B$)

❖ Conclusion: Therefore, it did not rain. ($\neg A$)

$$\frac{A \rightarrow B \quad \neg B}{\neg A} MT$$

Natural Deductions Proofs

❖ Show $p \rightarrow q, \neg q \vdash \neg p$

Line	Formula	Reason
1	$p \rightarrow q$	Premise
2	$\neg q$	Premise
3	$\neg p$	MT 1,2

Natural Deductions Proofs

❖ Implication Introduction ($\rightarrow I$)

❖ Implication Introduction is the rule that lets you prove a statement of the form $A \rightarrow B$.

❖ The idea is:

❖ Assume A , prove B , then conclude $A \rightarrow B$

❖ Assume its raining

❖ Show that the street is wet.

❖ Conclude: if its rains then street is wet.

$$\frac{\begin{array}{c} [A] \\ \vdots \\ B \end{array}}{A \rightarrow B} (\rightarrow I)$$

❖ Example:

❖ Show: $p \vdash q \rightarrow (p \wedge q)$

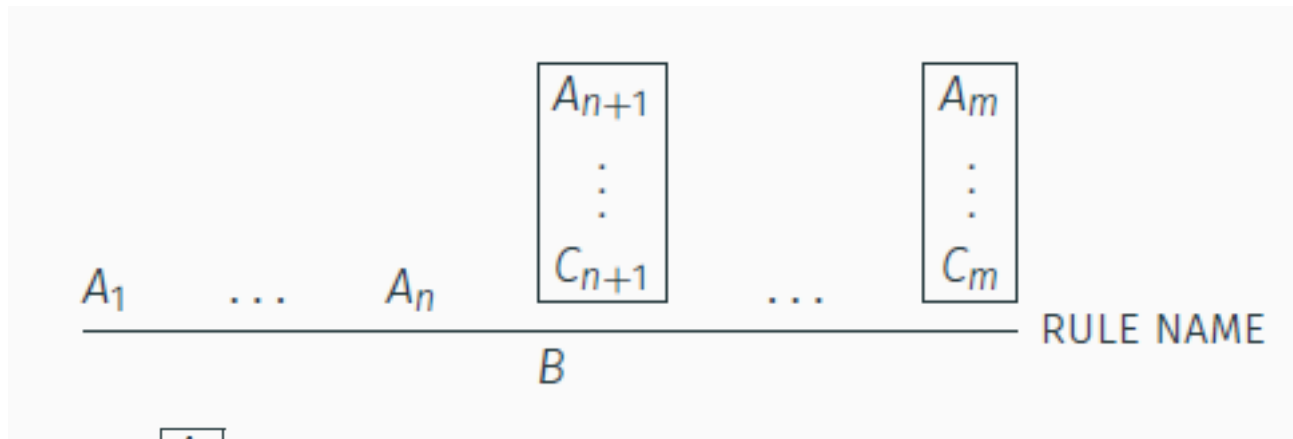
❖ Proof:

❖	1.	p	Premise
❖	2.	q	Assumption
❖	3.	$p \wedge q$	$\wedge I$ (1,2)
❖	4.	$q \rightarrow (p \wedge q)$	$\rightarrow I$ (2,3)

Natural Deductions Proofs

❖ Complex Inference Rule

- ❖ “given proofs and formulas derive a formula”.



- ❖ Meaning of $\boxed{\begin{array}{c} A_i \\ \vdots \\ C_i \end{array}}$ “assume A_i has been established in your proof and prove C_i .”

Natural Deductions Proofs

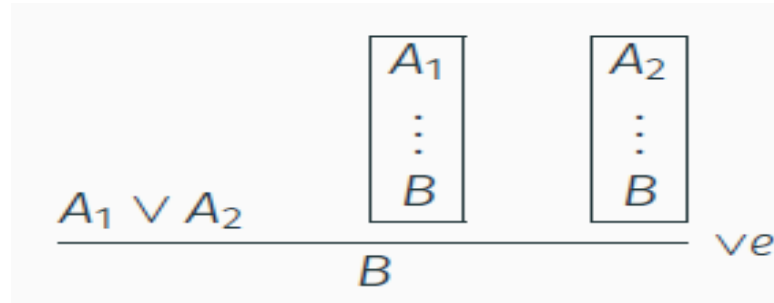
❖ Disjunction Introduction(Vi)

$$\frac{A1}{A1 \vee A2} \vee i1$$

$$\frac{A2}{A1 \vee A2} \vee i2$$

❖ Disjunction Elimination (Ve)

$$\frac{A1 \vee A2 \quad [A1] \vdash A3 \quad [A2] \vdash A3}{A3} \vee e$$



Example: $p \vee q, p \rightarrow r, q \rightarrow r \vdash r$.

- | | | |
|----|-------------------|-----------------------------|
| 1. | $(p \vee q)$ | premise |
| 2. | $p \rightarrow r$ | premise |
| 3. | $q \rightarrow r$ | premise |
| 4. | Assume p | |
| 5. | r | $\rightarrow e$ line 2,4 |
| 6. | Assume q | |
| 7. | r | $\rightarrow e$ line 3,6 |
| 8. | r | $\vee e$ line 1,4 – 5,6 – 7 |

Natural Deductions Proofs

❖ Prove: $p \vee q \vdash q \vee p$

❖ Proof

1.	$(p \vee q)$	premise
2.	p	Assumption
3.	$q \vee p$	$\vee i2$ line 2
4.	q	<i>Assumption</i>
5.	$q \vee p$	$\vee i1$ line 4
6.	$q \vee p$	$\vee e$, Line 1, 2-3; 4-5

$$\frac{p \vee q \quad [p] : q \vee p \quad [q] : q \vee p}{q \vee p} (\vee e)$$

Natural Deductions Proofs

❖ Prove: $(p \vee q) \vee r \vdash p \vee (q \vee r)$

1.	$(p \vee q) \vee r$	Premise
2.	$p \vee q$	Assumption
3.	p	Assumption
4.	$q \vee r$	$\vee I1, 3$
5.	$p \vee (q \vee r)$	$\vee I1, 4$
6.	q	Assumption
7.	$q \vee r$	$\vee I1, 6$
8.	$p \vee (q \vee r)$	$\vee I2, 7$
9.	$p \vee (q \vee r)$	$\vee E, 2, 3-5, 6-8$
10.	r	Assumption
11.	$q \vee r$	$\vee I2, 10$
12.	$p \vee (q \vee r)$	$\vee I2, 11$
13.	$p \vee (q \vee r)$	$\vee E, 1, 2-9, 10-12$

Contradictions

- ❖ Writing contradictions in the logic:
 - ❖ $p \wedge \neg p$
 - ❖ $(p \wedge q) \wedge \neg (p \wedge q)$
 - ❖ $(p \rightarrow \neg q \vee r) \wedge \neg (p \rightarrow \neg q \vee r)$
 - ❖ ...
- ❖ Contradictions are formulas whose semantics returns ***F*** for all models.
- ❖ They are *unsatisfiable*.
- ❖ Formulas of the form $A \wedge \neg A$ are unsatisfiable.

Contradictions

- ❖ They are all semantically equivalent: $A \wedge \neg A \equiv B \wedge \neg B$, for all A, B .
- ❖ Why?

A	$\neg A$	$A \wedge \neg A$
T	F	F
F	T	F

B	$\neg B$	$B \wedge \neg B$
T	F	F
F	T	F

- ❖ Since they have the same truth value in every interpretation, they are semantically equivalent.
- ❖ We should be able to prove $A \wedge \neg A \dashv\vdash B \wedge \neg B$, for all $A; B$.
 - ❖ $A \dashv\vdash B$ means $A \vdash B$ and $B \vdash A$

Contradictions

❖ How $A \wedge \neg A \vdash B$?

1. $A \wedge \neg A$	Premise
2. A	$\wedge E1$ 1
3. $\neg A$	$\wedge E2$ 1
4. $\perp$	2,3
5. B	$\perp E$ 4

❖ The symbol “ $\perp$ ” represents contradiction.

❖ The symbol “ $\perp$ ” (pronounced “bottom”) to represent $p \wedge \neg p$

❖ The symbol “ $\top$ ” (pronounced “top”) to represent : $\neg(p \wedge \neg p)$

❖ **Negation:**

$$\frac{A \quad \neg A}{\perp} \neg e$$

$$\frac{[A] \vdash \perp}{\neg A} \neg i$$

❖ Natural Deduction has a rule called *Explosion/Elimination ($\perp e$)*

$$\frac{\perp}{B} \perp e$$

❖ Intuition: if something as absurd as $A \wedge \neg A$ is considered *true* then any B can be shown to be *true*.

Contradictions



Prove: Show $p \rightarrow q, \neg q \vdash \neg p$.

1.	$p \rightarrow q$	premise
2.	$\neg q$	Premise
3.	p	Assumption
4.	q	$\rightarrow e$ Line 1,3
5.	$\perp$	$\neg e$ Line 2,4
6.	$\neg p$	$\neg i$ Line 3–5

Contradictions

❖ Prove: $(p \rightarrow (\neg q \vee r)) \wedge \neg(p \rightarrow (\neg q \vee r)) \vdash s$.

1.	$(p \rightarrow (\neg q \vee r)) \wedge \neg(p \rightarrow (\neg q \vee r))$	premise
2.	$(p \rightarrow (\neg q \vee r))$	$\wedge e1$ Line 1
3.	$\neg(p \rightarrow (\neg q \vee r))$	$\wedge e2$ Line 1
4.	$\perp$	$\neg e$ Line 2,3
5.	s	$\perp e$

❖ Therefore

$(p \rightarrow (\neg q \vee r)) \wedge \neg(p \rightarrow (\neg q \vee r)) \vdash s$

Contradictions



Prove: $p, \neg q \vdash \neg(p \rightarrow q)$

- | | | |
|----|-------------------------|--------------------------|
| 1. | p | premise |
| 2. | $\neg q$ | premise |
| 3. | $p \rightarrow q$ | assumption |
| 4. | q | $\rightarrow$ e Line 1,3 |
| 4. | $\perp$ | $\neg$ e Line 2,4 |
| 5. | $\neg(p \rightarrow q)$ | $\neg$ i Line3–5 |